

**NMAM INSTITUTE OF TECHNOLOGY, NITTE**  
**Off-Campus Centre of Nitte (Deemed to be University)**  
**First Semester B.Tech (CBCS) Degree Examinations**  
 December 2023

**EC1001-1 – BASIC ELECTRONICS**  
 (For AI&DS, AI&ML, CB, CCE, CSE, ISE, R&AI)

Duration: 3 Hours

Max. Marks: 100

**Note:**

**Part – A: Multiple Choice Questions:** Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

**Part – B: Descriptive Answer Questions:** Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

**PART - A: MULTIPLE CHOICE QUESTIONS****20 Marks**

- The reverse saturation current is more in \_\_\_\_\_ diode  
 A) Silicon diode B) Germanium diode  
 C) Zener diode D) Tunnel diode
- A rectifier is used to convert  
 A) DC to AC B) AC to DC  
 C) DC to DC D) AC to AC
- The maximum efficiency of a half wave rectifier circuit is  
 A) 38.2% B) 40.6%  
 C) 53.5% D) 100 %
- A bridge rectifier circuit is supplied through the transformer secondary voltage of  $100 \sin \omega t$  volts. The value of  $V_m$  is  
 A) 50 volts B) 70.7 volts  
 C) 100 volts D) 80 volts
- An voltage regulator designed using 14 V zener diode is supplied by a 20 volts battery. If a series resistance of  $1 \text{ k}\Omega$  is used, the current through the zener will be  
 A) 3 mA B) 2 mA  
 C) 6 mA D) 0 mA
- A BJT has  $\alpha = 0.99$ , the base current is  $100 \mu\text{A}$ . The value of collector current will be  
 A) 9.9 mA B) 100 mA  
 C) 10 mA D) 5 mA
- In a bipolar junction transistor, the collector current is controlled by  
 A) Collector voltage B) Emitter voltage  
 C) Collector resistance D) Base current
- If a 2 mV input signal produces a 2 V output, what is the voltage gain?  
 A) 0.001 B) 100  
 C) 0.004 D) 1000
- Amplifiers and oscillators are designed by operating the BJT in ..... region  
 A) Active B) Saturation  
 C) Cut off D) Inverted mode
- A bipolar junction transistor is a  
 A) Voltage controlled device B) Resistance controlled device  
 C) Current controlled device D) Impedance controlled device
- An ideal OPAMP has an input impedance of  
 A) zero B) 1  
 C) infinity D) -1
- An OPAMP used in closed loop configuration will have  
 A) Reduced gain B) Good stability  
 C) Increased bandwidth D) All of the above

13. A input signal of 1 volt is applied to the input of a voltage follower. The output voltage will be  
 A) 100 volts B) 10 volts  
 C) 50 volts D) 1 volt
14. In a junction field effect transistors, when drain voltage equals the pinch off voltage, the drain current  
 A) Increases with increase in drain voltage B) Remains constant with increase in drain voltage  
 C) decreases with increase in drain voltage D) Fluctuates with increase in drain voltage
15. JFET is a  
 A) Two terminal device B) Voltage controlled device  
 C) Current controlled device D) None of these
16. Choose the current statement when  $V_{GS} = 0V$  and  $V_{DS} = 0V$  in a JFET  
 A) The depletion regions around the pn junctions are equal in thickness and symmetrical B)  $I_D$  is maximum  
 C) The depletion regions around the pn junctions are not equal in thickness D) No change observed
17. An FET has  
 A) High input impedance B) Zero input impedance  
 C) Low input impedance D) None of the above
18. In a communication system, the noise can be generated at the  
 A) Transmitter, channel, receiver B) Channel, receiver  
 C) Transmitter, channel D) Only in the channel
19. Best example for natural noise is  
 A) Rain B) Traffic noise  
 C) Industrial noise D) Sound pollution
20. The speech signal frequency range is  
 A) 20 Hz to 20 kHz B) 30 to 34 kHz  
 C) 300 Hz to 3.4 kHz D) 10 to 10 kHz

### PART - B: DESCRIPTIVE ANSWER QUESTIONS

#### Unit – I

1. a) For the NPN silicon transistor connected in common base configuration, sketch and explain input and output characteristics.  
 b) Explain the forward and reverse characteristics of a Zener diode. Determine the value of series resistance,  $R_S$  in the circuit shown in Figure Q1(b) below. The voltage across the zener is 14 V and the zener current is 20 mA.

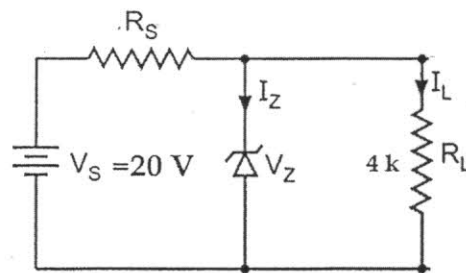


Figure Q.1(b)

2. a) With reference to a Full Wave Bridge Rectifier:  
 i) Draw the circuit diagram,  
 ii) Sketch the waveforms for input AC voltage, load current, load voltage,  
 iii) Explain the operation,  
 iv) Derive the expression for average DC load current and average DC load voltage.  
 b) With neat diagrams explain the construction, operation and drain characteristics of a N channel enhancement type MOSFET.

|       | Marks | BT* | CO* | PO* |
|-------|-------|-----|-----|-----|
| 1. a) | 8     | L2  | 2   | 1   |
| 1. b) | 8     | L3  | 1   | 1   |
| 2. a) | 8     | L2  | 1   | 1   |
| 2. b) | 8     | L2  | 2   | 1   |

3. a) Calculate the diode forward current  $I_F$  for the following diode circuits shown in i) Figure Q3a(i) and ii) Figure Q3a(ii).

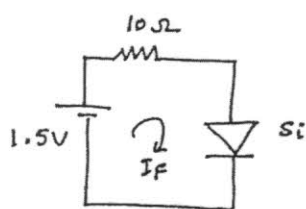


Figure Q3a(i)

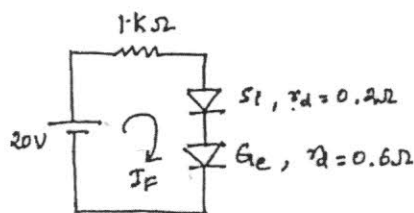


Figure Q3a(ii)

- b) With neat diagrams explain the construction, operation and drain characteristics of a n channel JFET.

|    |    |   |   |
|----|----|---|---|
| 6  | L3 | 1 | 1 |
| 10 | L2 | 2 | 1 |

### Unit – II

4. a) Draw the neat diagram for the Colpitt's oscillator circuit. In a Colpitt's oscillator,  $C_1 = C_2 = 400 \text{ pF}$  and  $L = 2 \text{ mH}$ . Calculate the frequency of oscillations.
- b) Explain the voltage series negative feedback with neat diagram and derive the expression for the closed loop gain.
- c) Design a non-inverting amplifier using OP-AMP with a closed loop gain of 21. Assume  $R_1 = 2 \text{ k}\Omega$ . Also draw the neat circuit diagram.
5. a) Explain the working of a Op-amp inverting amplifier with neat circuit diagram and waveforms. Also derive the expression for closed loop gain.
- b) In an oscillator circuit,  $L_1 = 10 \text{ mH}$ ,  $L_2 = 20 \text{ mH}$  and  $C = 0.01 \text{ }\mu\text{F}$ . Calculate: (i) Frequency of oscillations (ii) Feedback factor (iii) Gain (iv) Draw the circuit diagram
- c) Explain the voltage follower circuit using Op-amp with neat circuit diagram and necessary waveforms.
6. a) Design an adder circuit using OPAMP to generate an output voltage given by  $V_0 = -(4 V_1 + 2 V_2 + 6 V_3)$ . Chose  $R_f = 10 \text{ k}\Omega$ .
- b) Explain the operation of Astable multivibrator constructed using IC 555 Timer with necessary diagrams and waveforms.
- c) In an IC 555 Timer,  $R_1 = R_2 = 7.5 \text{ k}\Omega$  and  $C = 0.1 \text{ }\mu\text{F}$ . Calculate the frequency of oscillations.

|   |    |   |   |
|---|----|---|---|
| 6 | L3 | 3 | 1 |
| 6 | L2 | 4 | 1 |
| 4 | L3 | 3 | 1 |
| 6 | L2 | 4 | 1 |
| 6 | L2 | 3 | 1 |
| 4 | L2 | 4 | 1 |
| 6 | L3 | 4 | 1 |
| 6 | L3 | 3 | 1 |
| 4 | L2 | 4 | 1 |

### Unit – III

7. a) With a neat diagram, explain the basic elements of an electronic communication system.
- b) With neat diagrams, explain the elements of embedded systems.
- c) Explain the concept of frequency reuse in cellular communication system.
8. a) Discuss the types of channels used in a communication system with suitable examples.
- b) Compare the microprocessor and microcontroller.
- c) What is modulation? What are the different types of modulation schemes used commonly in a communication system? Explain briefly.

|   |    |   |   |
|---|----|---|---|
| 6 | L2 | 5 | 1 |
| 6 | L2 | 5 | 1 |
| 4 | L2 | 5 | 1 |
| 6 | L2 | 5 | 1 |
| 6 | L2 | 5 | 1 |
| 4 | L2 | 5 | 1 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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